WORKED SOLUTION

STEP II 2019, Question 7

Pure • Vectors and Scalar Products

STEP II 2019, Question 7

Read the question before you go on. It is in the STEP II paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

The key idea

For unit vectors, lengths and angles both come from scalar products: $|\mathbf{u}-\mathbf{v}|^2 = 2-2\,\mathbf{u}\cdot\mathbf{v}$. So the trick is to split the zero-sum condition, with some vectors on one side and the rest on the other, and take the length squared of each side. That turns the vector equation into equations between scalar products, and those tell us the shape.

Solution

Throughout we use the fact that for unit vectors $\mathbf{u}$, $\mathbf{v}$,

$$|\mathbf{u}\pm\mathbf{v}|^2 = \mathbf{u}\cdot\mathbf{u} \pm 2\,\mathbf{u}\cdot\mathbf{v} + \mathbf{v}\cdot\mathbf{v} = 2 \pm 2\,\mathbf{u}\cdot\mathbf{v}.$$

Part (i)

Since $\mathbf{a}+\mathbf{b} = -\mathbf{c}$, taking the length squared of both sides gives

$$2 + 2\,\mathbf{a}\cdot\mathbf{b} = |\mathbf{a}+\mathbf{b}|^2 = |\mathbf{c}|^2 = 1,$$

so $\mathbf{a}\cdot\mathbf{b} = -\frac{1}{2}$, as required.

The condition is symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$, so in the same way (writing $\mathbf{b}+\mathbf{c} = -\mathbf{a}$ and $\mathbf{c}+\mathbf{a} = -\mathbf{b}$) we get $\mathbf{b}\cdot\mathbf{c} = \mathbf{c}\cdot\mathbf{a} = -\frac{1}{2}$. Hence

$$AB^2 = |\mathbf{a}-\mathbf{b}|^2 = 2 - 2\left(-\tfrac{1}{2}\right) = 3,$$

and likewise $BC^2 = CA^2 = 3$. All three sides have length $\sqrt{3}$, so **triangle ABC is equilateral**. (Each pair of position vectors makes an angle of $120°$, and O is both the circumcentre and the centroid of the triangle, since $\frac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = \mathbf{0}$.)

Part (ii)

Write the condition as $\mathbf{a}_1 + \mathbf{a}_2 = -(\mathbf{a}_3 + \mathbf{a}_4)$ and take the length squared of both sides:

$$2 + 2\,\mathbf{a}_1\cdot\mathbf{a}_2 = 2 + 2\,\mathbf{a}_3\cdot\mathbf{a}_4 \quad\Longrightarrow\quad \mathbf{a}_1\cdot\mathbf{a}_2 = \mathbf{a}_3\cdot\mathbf{a}_4.$$

Grouping the vectors in the other two possible ways gives, in the same manner,

$$\mathbf{a}_1 \cdot \mathbf{a}_3 = \mathbf{a}_2 \cdot \mathbf{a}_4, \qquad \mathbf{a}_1 \cdot \mathbf{a}_4 = \mathbf{a}_2 \cdot \mathbf{a}_3.$$

Since $A_iA_j^2 = 2 - 2\,\mathbf{a}_i \cdot \mathbf{a}_j$, this says that *opposite edges are equal*: $A_1A_2 = A_3A_4$, $A_1A_3 = A_2A_4$ and $A_1A_4 = A_2A_3$.

(a) The four points in a plane

? **Ask yourself** Does the plane of the four points have to pass through the origin?

Yes. The centroid of the four points is $\frac{1}{4}(\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4) = \mathbf{0}$, and the centroid of points in a plane lies in that plane. So the plane contains O, and since every A_i is at distance 1 from O, the four points lie on the circle of radius 1 centred at O in that plane. From now on we work in this plane.

Let $\mathbf{w} = \mathbf{a}_1 + \mathbf{a}_2$, so that $\mathbf{a}_3 + \mathbf{a}_4 = -\mathbf{w}$.

If $\mathbf{w} = \mathbf{0}$, then $\mathbf{a}_2 = -\mathbf{a}_1$ and $\mathbf{a}_4 = -\mathbf{a}_3$.

If $\mathbf{w} \neq \mathbf{0}$, dot $\mathbf{a}_3$ with $\mathbf{a}_3 + \mathbf{a}_4 = -\mathbf{w}$ and use (ii):

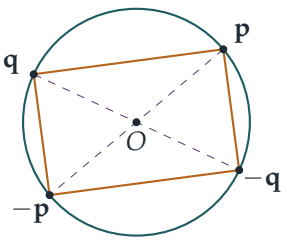
$$\mathbf{a}_3 \cdot \mathbf{w} = -(1 + \mathbf{a}_3 \cdot \mathbf{a}_4) = -(1 + \mathbf{a}_1 \cdot \mathbf{a}_2) = -\tfrac{1}{2}|\mathbf{w}|^2.$$

So $\mathbf{a}_3$ is a unit vector with a fixed component along $\mathbf{w}$: its tip lies on a fixed line perpendicular to $\mathbf{w}$, and a line meets a circle in at most two points. Now $-\mathbf{a}_1$ and $-\mathbf{a}_2$ are two such vectors, as

$$(-\mathbf{a}_1) \cdot \mathbf{w} = -(1 + \mathbf{a}_1 \cdot \mathbf{a}_2) = -\tfrac{1}{2}|\mathbf{w}|^2,$$

and similarly for $-\mathbf{a}_2$; they are different because $A_1 \neq A_2$. Hence $\mathbf{a}_3 = -\mathbf{a}_1$ or $\mathbf{a}_3 = -\mathbf{a}_2$, and then $\mathbf{a}_4 = -\mathbf{w} - \mathbf{a}_3$ is the other one.

In every case the four position vectors are $\mathbf{p}, -\mathbf{p}, \mathbf{q}, -\mathbf{q}$ for some unit vectors $\mathbf{q} \neq \pm\mathbf{p}$. So the four points are the ends of two diameters of the circle. The quadrilateral they form has diagonals that bisect each other (at O), so it is a parallelogram, and its diagonals are equal (both of length 2), so it is a **rectangle**. It is a square exactly when the two diameters are perpendicular.



(Strictly, the vertices should be taken in order around the circle. If the labels happen to go $A_1 = \mathbf{p}$, $A_2 = \mathbf{q}$, $A_3 = -\mathbf{q}$, $A_4 = -\mathbf{p}$, joining them in that order traces two sides and both diagonals of this rectangle.)

(b) A regular tetrahedron Now all six edges A_iA_j are equal, so all six scalar products $\mathbf{a}_i \cdot \mathbf{a}_j$ ($i \neq j$) take the same value, k say. Dotting $\mathbf{a}_1$ with the zero sum,

$$0 = \mathbf{a}_1 \cdot (\mathbf{a}_1 + \mathbf{a}_2 + \mathbf{a}_3 + \mathbf{a}_4) = 1 + 3k,$$

so $k = -\frac{1}{3}$. Then each edge has

$$A_1A_2^2 = 2 - 2k = 2 + \tfrac{2}{3} = \tfrac{8}{3}, \qquad \text{so the side length is} \qquad \sqrt{\tfrac{8}{3}} = \frac{2\sqrt{6}}{3}.$$

Checks

- The tetrahedron with vertices $\frac{1}{\sqrt{3}}(1,1,1)$, $\frac{1}{\sqrt{3}}(1,-1,-1)$, $\frac{1}{\sqrt{3}}(-1,1,-1)$, $\frac{1}{\sqrt{3}}(-1,-1,1)$ is regular, has unit position vectors summing to $\mathbf{0}$, and has edge $\frac{1}{\sqrt{3}}|(0,2,2)| = \frac{2\sqrt{2}}{\sqrt{3}} = \frac{2\sqrt{6}}{3} \approx 1.633$.
- Compare with (i): three unit vectors at 120° give side $\sqrt{3} \approx 1.732$, slightly longer, which makes sense since the tetrahedron's vertices are "more crowded" on the sphere.
- A computer check on thousands of random sets of four unit vectors with zero sum confirmed all three equalities of scalar products in (ii), and that in the planar case the points always come in opposite pairs $\pm\mathbf{p}$, $\pm\mathbf{q}$.

Common slips

- In (i), stopping at $\mathbf{a} \cdot \mathbf{b} = -\frac{1}{2}$: you need all three scalar products (or all three side lengths) before concluding that the triangle is equilateral.
- In (a), concluding "parallelogram" from equal opposite sides without checking how the vertices are arranged, or forgetting to show that the plane passes through O.

Beyond STEP

Sets of unit vectors summing to zero with all pairwise scalar products equal are the vertices of a *regular simplex*: in n dimensions there are $n+1$ of them, with $\mathbf{a}_i \cdot \mathbf{a}_j = -\frac{1}{n}$, which gives $-\frac{1}{2}$ and $-\frac{1}{3}$ above. Such configurations appear in chemistry (the 109.5° bond angle of methane is $\cos^{-1}(-\frac{1}{3})$) and in coding theory. The matrix of all scalar products $\mathbf{a}_i \cdot \mathbf{a}_j$, called the *Gram matrix*, determines a configuration of vectors up to rotation, which is the idea behind this whole question. Strang's *Introduction to Linear Algebra* is a good place to read more.

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